Superstore Sales Analysis Using SQL

SQL Project Presentation:

Analysis of Sales, Profit and Customer Trends Using SQL

Presented By:

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QUERY 1: Dataset Overview (Row Count and Distinct Values)

Query:

```
SELECT
COUNT(*) AS 'TOTAL ROWS',
COUNT(DISTINCT SHIPPING_MODE) AS 'SHIPPING_MODE',
COUNT(DISTINCT SEGMENT) AS 'SEGMENTS',
COUNT(DISTINCT CITY) AS 'CITIES',
COUNT(DISTINCT STATE) AS 'STATES',
COUNT(DISTINCT SUB_CATEGORY) AS 'SUB_CATEGORIES'
FROM SUPERSTORE;
```

Output:

TOTAL ROWS	SHIPPING_MODE	SEGMENTS	CITIES	STATES	SUB_CATEGORIES
9994	4	3	531	49	17

QUERY 2: Total Sales and Total Profit

Query:

```
SELECT SUM(SALES) 'TOTAL_SALES', SUM(PROFIT) 'TOTAL_PROFITS'
FROM SUPERSTORE;
```

Output:

TOTAL_SALES	TOTAL_PROFITS
2297200.860299955	286397.0217000013

QUERY 3: Analysis of Quantity (Average, Minimum, Maximum)

Query:

```
SELECT AVG(QUANTITY) 'AVERAGE_QUANTITY',MIN(QUANTITY) 'MIN_QUANTITY’,
MAX(QUANTITY) 'MAX_QUANTITY'
FROM SUPERSTORE;
```

Output:

AVERAGE_QUANTITY	MIN_QUANTITY	MAX_QUANTITY
3.7896	1	14

QUERY 4: Total Sales by Segment

Query:

```
SELECT SEGMENT,SUM(SALES) 'SALES_TOTAL'
FROM SUPERSTORE
GROUP BY SEGMENT
ORDER BY SUM(SALES) DESC;
```

Output:

SEGMENT	SALES_TOTAL
Consumer	1161401.3449999888
Corporate	706146.3668000001
Home Office	429653.1485000003

QUERY 5: Total Profit by Category

Query:

```
SELECT CATEGORY, SUM(PROFIT) 'TOTAL_PROFIT'
FROM SUPERSTORE
GROUP BY CATEGORY
ORDER BY SUM(PROFIT) DESC;
```

Output:

CATEGORY	TOTAL_PROFIT
Technology	145454.9480999999
Office Supplies	122490.80080000011
Furniture	18451.2728

QUERY 6: Total Sales by Region

Query:

```
SELECT REGION, SUM(SALES) 'SALES_TOTAL'
FROM SUPERSTORE
GROUP BY REGION
ORDER BY SUM(SALES) DESC;
```

Output:

REGION	SALES_TOTAL
West	725457.8245000006
East	678781.2399999979
Central	501239.8908000005
South	391721.9050000003

QUERY 7: State with the Highest Sales

Query:

```
SELECT T.STATE,T.SALES_TOTAL
FROM (SELECT STATE, SUM(SALES) AS SALES_TOTAL
FROM SUPERSTORE
GROUP BY STATE) AS T
WHERE T.SALES_TOTAL = (SELECT MAX(Q.SALES_TOTAL)
FROM (SELECT STATE, SUM(SALES) AS SALES_TOTAL
FROM SUPERSTORE
GROUP BY STATE) AS Q);
```

Output:

STATE	SALES_TOTAL
California	457687.631500001

QUERY 8: States with Negative Profit

Query:

```
SELECT STATE, SUM(PROFIT) AS TOTAL_PROFIT
FROM SUPERSTORE
GROUP BY STATE
HAVING SUM(PROFIT) < 0
ORDER BY TOTAL_PROFIT;
```

Output:

STATE	TOTAL_PROFIT
Texas	-25729.3563
Ohio	-16971.376600000018
Pennsylvania	-15559.960300000013
Illinois	-12607.88699999998
North Carolina	-7490.912200000003
Colorado	-6527.857900000001
Tennessee	-5341.6936
Arizona	-3427.9246
Florida	-3399.3017
Oregon	-1190.4704999999992

QUERY 9: Top Sub-Category by Sales for Each Region

Query:

```
SELECT T.REGION, T.SUB_CATEGORY, T.SALES_TOTAL

FROM (SELECT REGION, SUB_CATEGORY, SUM(SALES) AS SALES_TOTAL FROM SUPERSTORE GROUP BY REGION,
SUB_CATEGORY) AS T

WHERE T.SALES_TOTAL = (SELECT MAX(U.SALES_TOTAL)

FROM (SELECT SUB_CATEGORY,SUM(SALES) AS SALES_TOTAL FROM SUPERSTORE WHERE REGION = T.REGION
GROUP BY SUB_CATEGORY) AS U )

ORDER BY T.REGION;
```

Output:

REGION	SUB_CATEGORY	SALES_TOTAL
Central	Chairs	85230.64599999995
East	Phones	100614.98200000003
South	Phones	58304.437999999995
West	Chairs	101781.32799999989

QUERY 10: Total Number of Loss Making Rows

Query:

```
SELECT COUNT(*) AS LOSS_ROWS
FROM SUPERSTORE
WHERE PROFIT < 0;
```

Output:

LOSS_ROWS
1871

QUERY 11: Average Discount by Shipping Mode

Query:

```
SELECT SHIPPING_MODE, AVG(DISCOUNT) AS AVG_DISCOUNT
FROM SUPERSTORE
GROUP BY SHIPPING_MODE
ORDER BY AVG_DISCOUNT DESC;
```

Output:

SHIPPING_MODE	AVG_DISCOUNT
First Class	0.16460988296488774
Standard Class	0.16002345844504628
Same Day	0.15239410681399693
Second Class	0.13889460154241506

Query 12: Sub-Categories with Negative Profit

Query:

```
SELECT SUB_CATEGORY, SUM(PROFIT) AS TOTAL_PROFIT
FROM SUPERSTORE
GROUP BY SUB_CATEGORY
HAVING SUM(PROFIT) < 0
ORDER BY SUM(PROFIT);
```

Output:

SUB_CATEGORY	TOTAL_PROFIT
Tables	-17725.481100000008
Bookcases	-3472.5559999999978
Supplies	-1189.0994999999984

QUERY 13: State with the Highest Total Profit

Query:

```
SELECT t.STATE, t.PROFIT_TOTAL
FROM (SELECT STATE, SUM(PROFIT) AS PROFIT_TOTAL
FROM SUPERSTORE
GROUP BY STATE) AS t
WHERE t.PROFIT_TOTAL = (SELECT MAX(x.PROFIT_TOTAL)
FROM (SELECT STATE, SUM(PROFIT) AS PROFIT_TOTAL
FROM SUPERSTORE
GROUP BY STATE
) AS x);
```

Output:

STATE	PROFIT_TOTAL
California	76381.38710000017

QUERY 14: Highest Sales City in Each Region

Query:

```
SELECT T.REGION, T.CITY, T.SALES_TOTAL
FROM (SELECT REGION, CITY, SUM(SALES) AS SALES_TOTAL FROM SUPERSTORE
GROUP BY REGION,CITY) AS T
WHERE T.SALES_TOTAL = (SELECT MAX(U.SALES_TOTAL)
FROM (SELECT CITY,SUM(SALES) AS SALES_TOTAL FROM SUPERSTORE
WHERE REGION = T.REGION GROUP BY CITY) AS U)
ORDER BY T.REGION;
```

Output:

REGION	CITY	SALES_TOTAL
Central	Houston	64504.76039999994
East	New York City	256368.161
South	Jacksonville	44713.183
West	Los Angeles	175851.341

QUERY 15: Average Sales Per Row (Overall)

Query:

```
SELECT SUM(SALES)/COUNT(*) AS AVG_SALES_PER_ROW
FROM SUPERSTORE;
```

Output:

AVG_SALES_PER_ROW
229.8580008304938

QUERY 16: Average Sales Per Row by Segment

Query:

```
SELECT T.SEGMENT, (T.SUM_SALES/T.ROWS_TOTAL) AS AVG_SALES
FROM (SELECT SEGMENT,SUM(SALES) AS SUM_SALES, COUNT(*) AS ROWS_TOTAL
FROM SUPERSTORE GROUP BY SEGMENT) AS T
ORDER BY AVG_SALES DESC;
```

Output:

SEGMENT	AVG_SALES
Home Office	240.97204066180612
Corporate	233.8233002649007
Consumer	223.73364380658617

QUERY 17: Average Profit Per Row by Category

Query:

```
SELECT T.CATEGORY, (T.SUM_PROFIT/T.ROWS_TOTAL) AS AVG_PROFIT
FROM (SELECT CATEGORY, SUM(PROFIT) AS SUM_PROFIT, COUNT(*) AS ROWS_TOTAL
FROM SUPERSTORE GROUP BY CATEGORY) AS T
ORDER BY AVG_PROFIT DESC;
```

Output:

CATEGORY	AVG_PROFIT
Technology	78.75200221981586
Office Supplies	20.327049585131117
Furniture	8.699327109853842

Query 18: Total Quantity Sold by Sub-Category

Query:

```
SELECT SUB_CATEGORY, SUM(QUANTITY) AS QTY_TOTAL
FROM SUPERSTORE
GROUP BY SUB_CATEGORY
ORDER BY QTY_TOTAL DESC;
```

Output:

SUB_CATEGORY	QTY_TOTAL
Binders	5974
Paper	5178
Furnishings	3563
Phones	3289
Storage	3158
Art	3000
Accessories	2976
Chairs	2356
Appliances	1729
Labels	1400
Tables	1241
Fasteners	914
Envelopes	906
Bookcases	868
Supplies	647
Machines	440
Copiers	234

Query 19: Sub-Category with Highest Total Quantity

Query:

```
SELECT T.SUB_CATEGORY,T.QTY_TOTAL
FROM (SELECT SUB_CATEGORY, SUM(QUANTITY) AS QTY_TOTAL
FROM SUPERSTORE GROUP BY SUB_CATEGORY) AS T
WHERE T.QTY_TOTAL = (SELECT MAX(U.QTY_TOTAL)
FROM (SELECT SUB_CATEGORY, SUM(QUANTITY) AS QTY_TOTAL
FROM SUPERSTORE GROUP BY SUB_CATEGORY) AS U);
```

Output:

SUB_CATEGORY	QTY_TOTAL
Binders	5974

QUERY 20: Count of Discounted Rows

Query:

```
SELECT COUNT(*) AS DISCOUNTED_ROWS
FROM SUPERSTORE
WHERE DISCOUNT > 0;
```

Output:

DISCOUNTED_ROWS
5196

QUERY 21: Share of Discounted Rows by Segment

Query:

```
SELECT t.SEGMENT, t.DISC_ROWS / t.ROWS_TOTAL AS DISCOUNT_SHARE,
t.ROWS_TOTAL, t.DISC_ROWS
FROM (SELECT SEGMENT, COUNT(*) AS ROWS_TOTAL,
SUM(DISCOUNT > 0) AS DISC_ROWS
FROM SUPERSTORE GROUP BY SEGMENT) AS t
ORDER BY DISCOUNT_SHARE DESC;
```

Output:

SEGMENT	DISCOUNT_SHARE	ROWS_TOTAL	DISC_ROWS
Corporate	0.5275	3020	1593
Consumer	0.5259	5191	2730
Home Office	0.4896	1783	873

QUERY 22: Region with the Highest Average Discount

Query:

```
SELECT T.REGION,T.AVG_DISCOUNT
FROM (SELECT REGION,AVG(DISCOUNT) AS AVG_DISCOUNT
FROM SUPERSTORE GROUP BY REGION) AS T
WHERE T.AVG_DISCOUNT = (SELECT MAX(U.AVG_DISCOUNT)
FROM (SELECT REGION,AVG(DISCOUNT) AS AVG_DISCOUNT
FROM SUPERSTORE GROUP BY REGION) AS U);
```

Output:

REGION	AVG_DISCOUNT
Central	0.2403529918209213

QUERY 23: Number of Sub-Categories Sold per State

Query:

```
SELECT STATE, COUNT(DISTINCT SUB_CATEGORY) AS DISC_SUBCAT
FROM SUPERSTORE GROUP BY STATE ORDER BY DISC_SUBCAT,STATE;
```

Output:

STATE	DISC_SUBCAT
Wyoming	1
West Virginia	3
North Dakota	4
District of Columbia	6
Maine	6
Montana	8
South Dakota	8
Idaho	9
Vermont	9
Iowa	10
South Carolina	10
Kansas	11
New Mexico	11
Nebraska	12
New Hampshire	12
Arkansas	13
Rhode Island	13
Connecticut	14
Mississippi	14

STATE	DISC_SUBCAT
Oklahoma	14
Utah	14
Wisconsin	14
Indiana	15
Louisiana	15
Massachusetts	15
Minnesota	15
Missouri	15
Nevada	15
Alabama	16
Arizona	16
Delaware	16
Maryland	16
New Jersey	16
Oregon	16
Tennessee	16
California	17
Colorado	17
Florida	17

STATE	DISC_SUBCAT
Georgia	17
Illinois	17
Kentucky	17
Michigan	17
New York	17
North Carolina	17
Ohio	17
Pennsylvania	17
Texas	17
Virginia	17
Washington	17

QUERY 24: State with Maximum Product Variety

Query:

```
SELECT T.STATE, T.DISC_SUBCAT
FROM (SELECT STATE, COUNT(DISTINCT SUB_CATEGORY) AS DISC_SUBCAT
FROM SUPERSTOREGROUP BY STATE) AS T
WHERE T.DISC_SUBCAT = (SELECT MAX(U.DISC_SUBCAT)
FROM (SELECT STATE, COUNT(DISTINCT SUB_CATEGORY) AS DISC_SUBCAT
FROM SUPERSTOREGROUP BY STATE ) AS U);
```

Output:

STATE	DISC_SUBCAT
California	17
Colorado	17
Florida	17
Georgia	17
Illinois	17
Kentucky	17
Michigan	17
New York	17
North Carolina	17
Ohio	17
Pennsylvania	17
Texas	17
Virginia	17
Washington	17

QUERY 25: Fraction of Rows That Are Loss Making

Query:

```
SELECT
(SELECT COUNT(*) FROM SUPERSTORE WHERE PROFIT < 0) AS LOSS_ROWS,
(SELECT COUNT(*) FROM SUPERSTORE) AS ROWS_TOTAL,
(SELECT COUNT(*) FROM SUPERSTORE WHERE PROFIT < 0) /
(SELECT COUNT(*) FROM SUPERSTORE) AS LOSS_SHARE;
```

Output:

LOSS_ROWS	ROWS_TOTAL	LOSS_SHARE
1871	9994	0.1872

THANK YOU

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